

RC-28 Material Delivery and Cure Requirements

1. Material Delivery Records

Delivery and cure requirements are source records for construction staging review. They do not authorize construction or traffic loading.

Record ID	Material or Activity	Supplied Date / Quantity	Requirement	Constructability Note
MD-001	Precast box culvert units	Delivery confirmed 2026-09-12 06:30	Eight units required for CL-001	Later than project brief target setting date 2026-09-10 and later than crane availability in File 005.
MD-002	Headwall concrete mix	7.2 m3 available 2026-09-12 12:00	Certificate required before placement	Certificate ID not supplied in this document.
MD-003	Joint sealant	9 rolls delivered 2026-09-11 15:00	Seven joints required	Delivery quantity appears sufficient.
MD-004	Granular bedding	42 tonnes delivered 2026-09-09 13:00	Moisture control required	Delivery date supports base preparation.
MD-005	Asphalt surface course	310.4 m2 equivalent ordered	Place only after backfill acceptance	Quantity aligns with daily paving estimate but not schedule total.

2. Cure and Loading Windows

Cure ID	Element	Requirement	Start Basis	Release Condition
CU-001	Cast-in-place headwalls	Minimum 48 hours before backfill against headwall faces	Concrete placement time	Backfill can start after 48 hours and inspection release.
CU-002	Cast-in-place headwalls	Minimum 72 hours before traffic loading over adjacent approach	Concrete placement time	Traffic loading can start after 72 hours and cure verification.
CU-003	Non-shrink grout	Minimum 12 hours before joint wrapping	Grout placement time	Joint wrap after grout set.
CU-004	Asphalt surface	Place only when air temperature is at least 7 C and surface is dry	Paving shift	Do not place during active rainfall.
CU-005	Temporary cold patch	Not acceptable as final surface course	Interim traffic condition	Permanent asphalt reinstatement required.

3. Earliest Release Calculations

Calculation ID	Scenario	Supplied Input	Calculated Earliest Time	Note
CT-001	Headwall backfill release	Concrete placed 2026-09-12 14:00 plus 48 hours	2026-09-14 14:00	Backfill before this time should be treated as a cure-window conflict.
CT-002	Traffic loading release	Concrete placed 2026-09-12 14:00 plus 72 hours	2026-09-15 14:00	Traffic loading before this time should be treated as a cure-window conflict.
CT-003	Joint wrap release	Grout placed 2026-09-12 16:00 plus 12 hours	2026-09-13 04:00	Night work restriction must also be checked.
CT-004	Paving weather release	Dry surface and >=7 C required	not supplied	Weather input is missing from the material record.

4. Missing Material Documentation

Gap ID	Related Record	Missing Field	Impact
MG-001	MD-002	concrete_certificate_id	Concrete placement documentation cannot be verified from supplied material record.
MG-002	CT-004	paving_weather_reading	Asphalt placement window cannot be confirmed from material file alone.
MG-003	MD-001	delivery_offload_duration	Crane and delivery occupancy duration cannot be confirmed from delivery file.